MILITARY SPECIFICATION

DIRECTION FINDER SET AN/TRD-15( )

(UNITS OF)

1. SCOPE

1.1 This specification covers one type of direction finder equipment, designated Direction Finder Set AN/TRD-15( ). (See 6.5)

1.2 *The units covered in detail by this specification are:*

Antenna AS-1882( )/GR
Amplifier, Electronic Control, AM-3545( )/TRD
Cabinet, Electrical Equipment CY-4873( )/TRD
Cable Assemblies, RF, CG-92G/U (75 FT and 110 FT)
Coupler, Rotary, RF, CU-1096( )/TRD
Detector, RF, RF-159( )/TRD
Drive, Goniometer-Rotary Coupler, TG-126( )/TRD
Goniometer, Electrical, GO-42( )/TRD
Indicator, Azimuth, IP-669( )/TRD
Panel, Power Distribution, SB-1747( )/TRD
Power Supply, PP-4482( )/TRD
Receiver, Radio, R-725( )/URR
Transmitter, Radio, T-279( )/UR

2. APPLICABLE DOCUMENTS:

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 5825

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SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, Wrapping
QQ-S-781	Steel Strapping, Flat
UU-T-111	Tape, Paper, Gummed (Sealing and Securing)
PPP-B-585	Box, Wood, Wirebound
PPP-B-591	Box, Fiberboard, Wood-Cleated
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-S-760	Strapping, Nonmetallic, and Connectors
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, For Carton Sealing
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

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MIL-E-75	Electron Tubes, Packaging, Packing and Container Marking for General Specification For
MIL-C-104	Crates, Wood, Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-V-173	Varnish, Moisture and Fungus-Resistant, For the Treatment of Communications, Electronic and Associated Electrical Equipment

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MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys, Spot and Seam
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible
MIL-M-10578	Metal Conditioner and Rust Remover
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-R-13947	Receiver, Radio (Radio Receiver R-390( )/URR)
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-B-43291	Boxes, Fiberboard, Corrugated, Double-Wall, Weather Resistant
MIL-E-55301	Electromagnetic Compatibility

STANDARDS

MILITARY

MIL-STD-22	Welded Joint Design
MIL-STD-105	Sampling Procedures for Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment

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MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MS 20003	Indicator, Humidity, Card, Three Spot, Impregnated Areas
MS 91527	Mounts, Vibration, Cup-Type, Classes A and B, (For use with Electronic Equipment in Aircraft)

DRAWINGS

SC-D-23249	Cable Assy, RF, CG-92( )/U (___FT___IN)
SC-A-46439	List of Accessories for Package Tester
SC-B-61578	Grounding to Chassis
SC-DL-131710	Transmitter, Radio T-279( )/UR
SC-DL-535705	Amplifier, Electronic Control AM-3545( )/TRD
SC-DL-535730	Goniometer, Electronic GO-42( )/TRD
SC-DL-535740	Cabinet, Electrical Equipment CY-4873( )/TRD
SM-B-535747	Cable Assy, RF, CG-92G/U (75 Ft and 110 Ft) Test Procedures
SM-B-535748	Indicator, Azimuth, IP-669( )/TRD Test Procedures
SM-B-535755	Detector, RF, RF-159( )/TRD and Power Supply PP-4482( )/TRD Test Procedures
SM-B-535757	Coupler, Rotary, RF, CU-1096( )/TRD Test Procedures
SM-B-535758	Receiver, Radio, R-725( )/URR Test Procedures
SM-B-535759	System AN/TRD-15( ) In-Plant Test Procedures
SC-DL-535766	Indicator, Azimuth IP-669( )/TRD
SC-DL-535787	Antenna AS-1882( )/GR
SC-DL-535790	Coupler, Rotary, RF, CU-1096( )/TRD
SC-DL-535810	Drive, Goniometer-Rotary Coupler TG-126( )/TRD
SM-C-535791	Coupler, Rotary, R.F. CU-1096/TRD Assembly

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SC-DL-535840	Power Supply PP-4482( )/TRD
SC-DL-535842	Panel, Power Distribution SB-1747( )/TRD
SC-DL-535880	Receiver, Radio R-725( )/URR
SM-B-535902	Plunger Assembly
SC-DL-535929	Detector, RF, RF-159( )/TRD
SM-B-535948	Panel, Power Distribution SB-1747( )/TRD Test Procedures
SM-B-535949	Amplifier, Electronic Control AM-3545( )/TRD Test Procedures
SM-B-536029	Drive, Goniometer-Rotary Coupler TG-126( )/TRD Test Procedures
SM-C-536043	Slide, Long, Assy (R.H. & L.H.)
SM-B-536123	Goniometer, Electrical, GO-42( )/TRD Test Procedures
SM-B-536137	Transmitter, Radio T-279( )/UR Test Procedures

3. REQUIREMENTS

3.1 Description.- Direction Finder Set AN/TRD-15( ) is a radio direction finder employing Doppler's principle which accurately detects and indicates, both visually and aurally, the azimuth bearing of CW, AM, FM and FSK signals over a frequency range of 1.5 to 20.0 mcs. The equipment consists of a series of 25 monopole antennas arranged equidistantly along the circumference of a 150-foot (or 300-foot) diameter circle plus a single monitor monopole antenna located near the center of the circle formed, an antenna rotary coupler, RF detector units, radio receivers and visual indicators. Components of the operating positions are housed in dual console cabinets; thus, simultaneous operation of two direction finding positions from a common antenna system is available. A target transmitter is used for system calibration.

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3.2 Preproduction samples.- The contractor shall furnish two preproduction (first article) samples of each unit on order for approval provided that the invitation for bids and the contract require samples. Prior to initiation of the Preproduction Tests, the contractor shall submit to the Contracting Officer's Technical Representative, three copies of the test data on the contractor's preliminary check tests performed on the preproduction samples. One of the approved preproduction samples shall be retained for use by the Government Quality Assurance Representative as a reference durint the life of the contract. Approved preproduction samples shall be refurbished as necessary to meet full specification requirements and shipped as part of the production quantity on order. (See 4.3)

3.3 Construction.- The following items shall be constructed in accordance with the requirements of this specification and of the indicated Electronics Command Drawings and Data Lists (including the requirements for parts, materials and processes thereon):

Antenna AS-1882( )/GR	SC-DL-535787
Amplifier, Electronic Control AM-3545( )/TRD	SC-DL-535705
Cabinet, Electrical Equipment CY-4873( )/TRD	SC-DL-535740
Cable Assembly, RF, CG-92G/U (75 FT)	SC-D-23249
Cable Assembly, RF, CG-92G/U (110 FT)	SC-D-23249
Coupler, Rotary, RF, CU-1096( )/TRD	SC-DL-535790
Detector, RF, RF-159( )/TRD	SC-DL-535929
Drive, Goniometer-Rotary Coupler, TG-126( )/TRD	SC-DL-535810
Goniometer, Electrical, GO-42( )/TRD	SC-DL-535730
Indicator, Azimuth, IP-669( )/TRD	SC-DL-535766
Panel, Power Distribution, SB-1747( )/TRD	SC-DL-535842
Power Supply, PP-4482( )/TRD	SC-DL-535840
Receiver, Radio, R-725( )/URR	SC-DL-535880
Transmitter, Radio, T-279( )/UR	SC-DL-131710

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3.4 Procurement model.- A procurement model of the equipment will be available for inspection by prospective bidders and will be lent to the contractor if specified in the invitation for bids and contract. Physical construction of the equipment shall conform to the drawings. In case of conflict between the specified performance characteristics and the performance of the model, the specified performance characteristics shall govern.

3.5 Parts, materials and processes (see 4.4).- In addition to the requirements of this specification, the requirements of Specification MIL-P-11268, including the selection requirements therein, shall apply.

3.5.1 Marking.- Marking shall conform to Specification MIL-M-13231. Front Panel marking shall be Group 1 as described in that specification.

3.5.2 Cleaning.-

3.5.2.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted.

3.5.2.2 Units.- After assembly the units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.5.3 Soldering.-

3.5.3.1 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used when they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

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3.5.3.2 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a large area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.5.4 Welding.- Wherever practicable, welded joints shall conform to Standard MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858.

3.5.4.1 Cleaning prior to welding.- Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.5.4.2 Process.- Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock-marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used, when practicable, for welding aluminum, magnesium, or stainless steel.

3.5.4.3 Cleaning after welding.- Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with Specification MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed.

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3.5.5 Connections, soldered, general.- Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to a fixed terminal shall be looped at least one-half turn but not more than three-quarter turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and nonmetallic material. Fraying of textile ends of wires shall be prevented mechanically or by the application of varnish conforming to Specification MIL-V-173. No varnsih, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.5.6 Grounding.- Ground connection to shields and to other mechanical parts, except the chassis or frame, shall not be made to complete electrical circuits but only to eliminate high-potential ac points.

3.5.6.1 Grounding to chassis.- Grounding to chassis shall conform to Drawing SC-B-61578.

3.5.7 Finish.- The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings.

3.6 Serial numbers.- The following shall be serial numbered:

Amplifier, Electronic Control, AM-3545( )/TRD
Coupler, Rotary, RF, CU-1096( )/TRD
Detector, RF, RF-159( )/TRD
Drive, Goniometer-Rotary Coupler, TG-126( )/TRD
Goniometer, Electrical, GO-42( )/TRD
Indicator, Azimuth, IP-669( )/TRD
Panel, Power Distribution, SB-1747( )/TRD
Power Supply, PP-4482( )/TRD
Receiver, Radio, R-725( )/URR
Transmitter, Radio, T-279( )/UR

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3.7 Interchangeability (see 4.7).- Like units, assemblies, subassemblies and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be handpicked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.8 Ancillary items.- Ancillary items are required to permit operation of Direction Finder Set AN/TRD-15 as a system. When procured, these items shall conform to the requirements stated in the bid request and contract. When required as test equipment in the procurement of items covered by this specification, the ancillary items will be lent to the contractor by the Government as specified in the bid request and contract.

3.9 Service conditions (see 4.8).- The equipment shall meet the following service conditions--where a test is referenced, meeting the tests shall be considered as compliance with the requirements. Unless otherwise specified, these requirements apply to complete systems and units listed in 1.2.

3.9.1 Temperature (see 4.8.1).-

3.9.1.1 Operating.- Ambient temperature in the range of +150° F to 0° F. (The 150° F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

3.9.1.2 Nonoperating.- Exposure in the range of +160° F to -80° F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.9.2 Moisture-Resistance (see 4.8.2).- Exposure up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation for 4 hours. Meeting the tests of 4.8.2 shall be considered compliance with this requirement.

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3.9.3 Elevation (see 4.8.3).-

Operating: Up to 10,000 feet above sea level.

Nonoperating: Up to 50,000 feet above sea level.

3.9.4 Rain (see 4.8.4) (Transmitter only).- Two inches of rainfall per hour, with wind at 20 miles per hour.

3.9.5 Vibration (Internal) (see 4.8.5) (applicable to 1.2 units only).- The amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second.

3.9.6 Shock (see 4.8.6) (applicable to 1.2 units only).-

3.9.6.1 Bounce, vehicular.- The equipment shall show no physical damage and shall meet specified performance after subjection to the Bounce test of 4.8.6.1.

3.9.6.2 Benchhandling (see 4.8.6.2).- The equipment shall show no physical damage and shall meet full specification performance after subjection to the tests of 4.8.6.2.

3.10 Preconditioning.- The equipment shall be capable of meeting the requirements herein after subjection to the bounce and electrical preconditioning of 4.6.

3.11 Electromagnetic compatibility (see 4.11) (Does not apply to Transmitter).- The equipment shall meet the emission and susceptibility limits of Specification MIL-E-55301 for tactical equipment. Application of suppression components and techniques shall conform to that specification.

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3.12 Detailed requirements (see 4.9).-

3.12.1 Input power (see 4.9.1)

3.12.1.1 The direction finder system (not including Transmitter) shall be capable of meeting the performance requirements of this specification when operated from any power source having a voltage of 115 volts ± 10 volts or 230 volts ± 20 volts and a frequency of 47 to 63 cps. The units listed in 1.2 shall be capable of meeting the performance requirements of this specification when operated from any power source having a voltage of 115 vac ± 10 volts and a frequency of 47 to 63 cps.

3.12.1.2 Transmitter, Radio T-279( )/UR shall be capable of meeting the performance requirements of this specification when operated from a dc source of ± 10% of the 1.5, 67.5, 90 and 135 volts.

3.12.2 Amplifier, Electronic Control, AM-3545( )/TRD (see 4.9.2).- The Amplifier, Electronic Control, AM-3545( )/TRD shall provide a constant 180 cps output with sufficient power output to operate the Drive, Goniometer-Rotary Coupler TG-126( )/TRD. A tuning fork is used to supply and maintain a constant supply frequency for the motor. This unit shall meet the following requirements:

3.12.2.1 Tuning fork period (see 4.9.2.1).- The tuning fork signal frequency period shall be 5552.77-5558.33 microseconds.

3.12.2.2 Synchronism (see 4.9.2.2).- The Amplifier shall be capable of operating the drive motor of Drive TG-126/TRD at synchronous speed.

3.12.2.3 Protective circuit (see 4.9.2.3).- The protective circuit shall function to protect the associated circuits when a locked-motor, short, or open condition occurs at the output.

3.12.2.4 Output voltage (see 4.9.2.4).- The output voltage of the Amplifier shall be 115 vac ± 10% when operating with a synchronous load.

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3.12.3 Cable Assembly, RF, CG-92G/U (75 Ft and 110 Ft) (see 4.9.3).- Antenna Cable Assemblies shall be cut in lengths to match the impedance and phase angle of a standard reference cable of 75 ft 6 in, or 110 ft 6 in, as the case may be, fabricated in accordance with ECOM Drawings SC-D-23249 and SM-B-535747.

3.12.4 Coupler, Rotary, RF, CU-1096( )/TRD (see 4.9.4).- The Coupler, Rotary, RF, CU-1096( )/TRD functions to simulate rotation of the antenna system by sampling the voltages generated in each of the 25 monopole antennas, and applies the signal to the input of Radio Receiver R-725( )/URR. This unit shall meet the following requirements:

3.12.4.1 Blending (see 4.9.4.1).- The commutation characteristics between any input connector and either output connector shall be such that a smooth bell shaped and symmetrical curve is obtained.

3.12.4.2 Blending amplitude modulation (see 4.9.4.2).- The blending amplitude variation at each antenna output shall not exceed 6%.

3.12.4.3 Output isolation (see 4.9.4.3).- Isolation between output receptacles shall exceed 24 db.

3.12.4.4 Insertion loss (4.9.4.4).- Insertion loss of output A (J26) and output B(J27) shall be no greater than specified below:

Frequency (mcs)	J26 and J27 Insertion Loss (db) Max.
1.5	13
3	11.5
5	11.5
10	14
15	17
20	18

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3.12.4.5 Rotor-Stator spacing (see 4.9.4.5).- The spacing between the rotor and stator shall comply with the gap specified on Drawing SM-C-535791.

3.12.5 Detector, RF, RF-159( )/TRD (see 4.9.5).- The RF Detector functions to remove any amplitude modulation present in the received signal and converts the phase modulation into a 90 cps output. This unit shall meet the following requirements when used with Power Supply PP-4482( )/TRD.

3.12.5.1 Loud speaker (see 4.9.5.1).- The speaker shall reproduce a test frequency of 1000 cps moderately loud and clear.

3.12.5.2 Filament voltage (see 4.9.5.2).- The filament supply voltage shall measure 5.9 to 6.5 vac with line voltage set to 115 vac.

3.12.5.3 Plate supply voltage (see 4.9.5.3).- The voltage at test point TP13 shall measure 250 ± 5.0 vdc with line voltage set to 115 vac.

3.12.5.4 Local oscillator (see 4.9.5.4).- The peak-to-peak oscillator output voltage observed on an oscilloscope, of test point TP1 shall measure 0.4 to 2.0 volts.

3.12.5.5 Receiver relay (see 4.9.5.5).- A switch operated relay shall be provided to select either the local oscillator output or the monitor receiver output.

3.12.5.6 AM limiting (see 4.9.5.6).- The amplitude variation observed on an oscilloscope at test point TP7 (and TP3) shall not exceed 1%.

3.12.5.7 Non symmetrical limiting (see 4.9.5.7).- The FM modulation observed on an oscilloscope at test point TP7 (and TP3) shall not exceed 5% of the length of one complete cycle.

3.12.5.8 Limiting characteristic (see 4.9.5.8).- The peak-to-peak level observed on an oscilloscope, at test point TP7 (and TP3) shall measure 15-20 volts.

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3.12.5.9 High-low level difference (see 4.9.5.9).- The difference between the voltage levels observed in 3.12.5.8 with high and low level inputs shall not exceed 0.2 volts.

3.12.5.10 5 kc level (see 4.9.5.10).- The peak-to-peak amplitude observed on an oscilloscope, at test point TP9, shall measure 0.4 - 2.0 volts.

3.12.5.11 Discriminator stability (see 4.9.5.11).- The peak-to-peak amplitude observed on an oscilloscope, at test point TP11, shall measure 5v ± 2v.

3.12.5.12 Discriminator linearity (see 4.9.5.12).- The oscilloscope pattern observed, at test point TP11, shall not exceed 0.5 division when measured per 4.9.5.12.

3.12.5.13 Auxiliary output, high (see 4.9.5.13).- The peak-to-peak amplitude observed on an oscilloscope, at test point TP12, shall measure 3.5 - 12 volts (frequency deviation 3 kcs).

3.12.5.14 Auxiliary output, low (see 4.9.5.14).- The peak-to-peak amplitude observed on an oscilloscope, at test point TP12, shall measure 0.3 - 3.5 volts (frequency deviation 1 kc).

3.12.5.15 Harmonic rejection (see 4.9.5.15).- The second, third and fourth harmonic voltage shall each be at least 30 db below the fundamental (90 cps) reference voltage.

3.12.6 Drive, Goniometer-Rotary Coupler, TG-126( )/TRD (see 4.9.6).- The Drive, Goniometer-Rotary Coupler TG-126( )/TRD functions to rotate two Goniometers, Electrical, GO-42( )/TRD in electrical synchronism. A synchronizing voltage generator is part of this unit. This unit shall meet the following requirements:

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3.12.6.1 Rotational synchronism (see 4.9.6.1).- The dual oscilloscope display of SYNC voltage generator pulse output (J1, GONIO Drive Chassis) and Amplifier, Electronic Control, AM-3545( )/TRD output (J2) shall be in continuous synchronism. The Gonio Drive Motor speed shall be 5400 rpm ± 2.7 rpm and the frequency period shall be 5552.77 - 5558.33 microseconds.

3.12.6.2 Goniometer orientation (see 4.9.6.2).- Each Goniometer housing shall be capable of ± 15° physical rotation on mount, with no evidence of binding.

3.12.6.3 Synchronizing pulse orientation (see 4.9.6.3).- The electrical phase relationship between the motor power signal (J2, Amplifier, Electronic Control, AM-3545( )/TRD) and the Synchronizing Pulse (J1, Gonio Dirve Chassis) shall be adjustable through 360°.

3.12.6.4 Synchronizing pulse width (see 4.9.6.4).- The synchronizing pulse width shall be less than 0.2 milliseconds.

3.12.6.5 Synchronizing pulse amplitude (see 4.9.6.5).- The synchronizing pulse amplitude shall not be less than 0.2 volts positive peak and 0.8 volts negative peak.

3.12.7 Goniometer, Electrical, GO-42( )/TRD (see 4.9.7).- The Goniometer, Electrical, GO-42( )/TRD provides reference signals required by the Indicator, Azimuth, IP-669( )/TRD visual cathode-ray tube display unit. The output of the Goniometer is a 90 cps amplitude modulated 72 kc carrier signal. This unit shall meet the following requirements:

3.12.7.1 Bearing accuracy and null ratio (see 4.9.7.1).- All direct and reciprocal bearing accuracies shall be ± 1° and null ratios a minimum of 45 db.

3.12.7.2 Balance ratio (see 4.9.7.2).- The balance ratio shall be a minimum of 40 db.

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3.12.7.3 Insertion loss (see 4.9.7.3).- The insertion loss shall be a maximum of 18 db.

3.12.8 Indicator, Azimuth, IP-669( )/TRD (see 4.9.8).- The Indicator, Azimuth, IP-669( )/TRD receives a signal, whose phase is determined by the wavefront approach to the antennas, amplifies and further processes it and presents a visual display of bearing. This unit shall meet the following requirements:

3.12.8.1 Sense and bearing display (see 4.9.8.1).- The visual cathode-ray tube display shall contain automatic sense. The pattern shall take the form of a half propeller-like display. Sense shall be correct and distinct at all frequencies. The circuit used to blank one half of the Indicator, RF cycle shall be operative at all times.

3.12.8.2 Indicator high voltage supply (see 4.9.8.2).- The high voltage supply shall develop at least 4000 vdc for operation of the cathode-ray tube and the Indicator rf sweep frequency shall be 72 kcs ± 0.1 kc.

3.12.8.3 Indicator low voltage supply (see 4.9.8.3).- With the low voltage supply output adjusted to 275 vdc at 115 vac line input, the output voltage shall not vary more than 1% when the line voltage is varied ± 10 volts.

3.12.8.4 Indicator pattern size and quality (see 4.9.8.4).- The output of the Indicator Goniometer shall be sufficient to produce a circular pattern of at least 6 inches in diameter under conditions of no signal input from the Radio Receiver R-725( )/URR. The circular pattern shall be clear and sharply focused and the component circles of the pattern shall appear perfectly superimposed. It shall be free from observable circumferential flattening or distortion. The centers of the component circles shall be perfectly superimposed and shall appear as a star or dot having a greatest dimension not exceeding 1/16 inch when the pattern diameter is 6 inches. The 1/2 propeller-shaped pattern quality produced by injection of the direction finding operating voltage from the Radio Frequency Detector RF-159( )/TRD shall have a sharp tip and shall be free from angular "splits", caused by factors within the Indicator including the Goniometer and its assembly. The sense pattern shall be sharply focused and free from "ghosts" due to incomplete blanking. There shall be less than 10% change of pattern size and no change of shape or positioning over the normal range of Intensity control adjustment.

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3.12.8.5 Indicator sensitivity (see 4.9.8.5).- The sensitivity of the Indicator shall be such that by injection of 90 cps generator output into the Indicator input (J1) it shall require less than 3 volts peak-to-peak to produce a full scale clear and undistorted 1/2-propeller display, on the cathode-ray tube.

3.12.8.6 Indicator accuracy (see 4.9.8.6).- The error introduced by the Indicator, including Goniometer, in measuring the phase of the Direction Finder operating voltage from Radio Receiver R-725( )/URR shall not exceed 1.5 degrees when measured at any phase angle from 0 to 360 degrees.

3.12.8.7 Indicator cathode-ray tube shielding (see 4.9.8.7).- The magnetic shield for the cathode-ray tube shall minimize extraneous beam deflection produced by steady or periodic external fields. It shall be constructed of a high permeability magnetic material, suitably annealed to produce low magnetic retentivity. The shield shall be of such thickness that the stray deflection produced by a homogeneous external field of 0.5 gauss normal to the cathode-ray tube axis is less than 1/8 inch.

3.12.8.8 Indicator alidade and azimuth scale assembly (see 4.9.8.8).- The Indicator shall be provided with an alidade and azimuth scale assembly fitted about the face of the cathode-ray tube. The alidade shall have two crossed hair lines and shall be capable of operating smoothly over 120 degrees of rotation. The hair lines shall have a length equal to the inside diameter of the azimuth scale. The accuracy of calibration of the alidade and azimuth scale assembly shall be such that the maximum error due to azimuth scale shall not exceed 1/4 degree; including the error due to azimuth scale distortion, the error in quadrature of the alidade crossed hair lines and the error incurred by eccentricity of the alidade with respect to the azimuth scale. The observational error due to parallax between the scale and the hair lines shall not exceed 1/4 degree.

3.12.8.9 Cathode-ray tube quadrature (see 4.9.8.9).- The perpendicularity between horizontal and vertical plates shall be 1/2 degree or better.

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3.12.9 Panel, Power Distribution, SB-1747( )/TRD (see 4.9.9.).- The Panel, Power Distribution, SB-1747( )/TRD is a 2 kw autotransformer which permits selecting and adjusting 115 vac or 230 vac input power to the 115 vac required for operation of the Direction Finder Set AN/TRD-15 system. This unit shall meet the following requirements:

3.12.9.1 Output meters (see 4.9.9.1).- The output voltmeter shall indicate the voltage applied to the load. The frequency meter shall indicate line frequency within the range 48 - 52 cps and 58 - 62 cps.

3.12.9.2 Output voltage range (see 4.9.9.2).- With 115 or 230 vac input, the output voltage with full load shall be capable of adjustment over the range of 105 volts to 125 volts.

3.12.9.3 Overload (see 4.9.9.3).- Circuit breakers in the 115 and 230 volt lines shall trip within 7 - 30 seconds (under ambient temperature of 70 ± 10° F) after a 4000 watt load is applied. The application of a 4000 watt load for 30 seconds shall not adversely affect the Autotransformer.

3.12.10 Power Supply PP-4482( )/TRD (see 4.9.10).- The Power Supply PP-4482( )/TRD provides regulated power to the Detector, RF, RF-159( )/TRD. This unit shall meet the following requirements:

3.12.10.1 B+ voltage (see 4.9.10.1).- The output voltage shall be 250 ± 2 vdc under conditions of voltage input variation from 105 to 125 vac and load variation from half load to full load.

3.12.10.2 Filament voltage (see 4.9.10.2).- Filament voltage output shall conform to the following:

Line volts (vac)	Load	Filament volts out (vac)
115	None	6.0 - 6.7
115	Full	5.8 - 6.5
125	Full	6.3 - 7.0
105	Full	5.1 - 5.9

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3.12.10.3 Hum and noise (see 4.9.10.3).- The hum and noise voltage measured at TP1 shall not exceed 0.25 volts peak-to-peak.

3.12.11 Receiver, Radio R-725( )/URR (see 4.9.11).- Receiver, Radio, R-725( )/URR is a modified Receiver R-390A/URR operating in the High Frequency range. R-390A/URR Receiver will be furnished by the Government for modification as specified in the bid request and contract. They shall be modified in accordance with ECOM Drawing SC-DL-535880 and shall meet the requirements of Specification MIL-R-13947 with the following additions and corrections:

3.12.11.1 IF phase response (see 4.9.11.1).- The phase shift shall be no more than 3.0 degrees from minus 750 cps to plus 750 cps of 455.000 kcs. The phase shift from the IF frequency (455.000 kcs) to either plus or minus 750 cps shall not exceed 2.0 degrees.

3.12.11.2 IF Selectivity (see 4.9.11.2).- The IF selectivity shall be as specified below:

Receiver Bandwidth Switch Position (kcs)	Bandwidth (db)	Specification Limits (kcs)
0.1	6.0	0.1 + 50% - 0
0.1	6.0	1.0 +50% - 0
	60.0	4.8 (Max)
2.0	6.0	2.0 ± 15%
	60.0	6.8 (Max)

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Receiver Bandwidth Switch Position (kcs)	Bandwidth (db)	Specification Limits (kcs)
4.0	6.0	4.0 ± 15%
	60.0	11.0 (Max)
8.0	3.0	7.0 (Min)
	6.0	11.5 (Max)
	60.0	22.0 (Max)
	Peak-To-Valley Ratio =	4.0 db (Max)
16.0	3.0	12.0 (Min)
	6.0	17.0 (Max)
	60.0	31.0 (Max)
	Peak-To-Valley Ratio =	4.0 db (Max)

3.12.11.3 Beat frequency oscillator leakage (see 4.9.11.3).- The oscillator leakage shall not exceed 350 uv.

3.12.11.4 IF output (see 4.9.11.4).- The IF output shall exceed 20 mv for 3 uv input signal. IF output shall measure 170 - 230 mv for 30 uv input signal.

3.12.11.5 AM sensitivity (see 4.9.11.5).- The (S+N)/N Ratio shall exceed 10 db for input frequencies and signal levels specified below:

Frequency (mcs)	Signal Level (UV)
0.5 to 16	3.3
16 to 32	4.4

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3.12.11.6 A.G.C. Characteristic (see 4.9.11.6).- The audio output shall not exceed the following audio output limits at a test frequency of 16 mcs.

Input Level (UV)	Audio Output (DB)
5	0 (5 MV Reference)
1,000	5
10,000	10
100,000	18

3.12.11.7 Overall gain (see 4.9.11.7).- The signal generator output shall measure 1.0 - 4.0 uv for 7.0 vdc diode load (TB-103 Term 14-15) at 4.0 and 16.0 mcs.

3.12.11.8 BFO tuning range (see 4.9.11.8).- With the beat frequency oscillator tuned to 455 kc, and the BFO PITCH control knob set at the vertical "0" position, rotation of the control in either direction to the panel marking "+3" or "-3" shall produce a change in frequency of not less than 2.4 kcs and not more than 3.6 kcs.

3.12.11.9 IF Frequency (see 4.9.11.9).- The IF frequency shall be 454.7 - 455.3 kcs (J116).

3.12.11.10 Carrier level meter (see 4.9.11.10).- The carrier level meter indication shall change 20 db ± 10 db for each 20 db input level change (20, 40, 60, 80 and 100 db), reference input level 50 uv for 20 db point).

3.12.12 Transmitter, Radio T-279( )/UR (see 4.9.12).- The Transmitter, Radio T-279( )/UR is a self-contained battery-powered portable unit intended for use as a signal source for calibration and checking the operation of the direction finder equipment in the field. This unit shall meet the following requirements:

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3.12.12.1 Frequency range, overlap and calibration accuracy (see 4.9.12.1).- The Transmitter shall operate over a frequency range of 0.5 - 30.0 mcs in six bands as follows, with each band overlapping the adjacent band. The indicated frequency shall be accurate to within 1%.

Frequency Band	Frequency Range (MCS)
1	0.5 - 1.0
2	1.0 - 2.0
3	2.0 - 4.0
4	4.0 - 8.0
5	8.0 - 16.0
6	16.0 - 30.0

3.12.12.2 Frequency drift (see 4.9.12.2).- Frequency drift shall not exceed 5 kcs.

3.12.12.3 Frequency of modulation (period) (see 4.9.12.3).- The modulation frequency period shall be 1.5 - 3.0 milliseconds.

3.12.12.4 Depth of modulation (negative peak) (see 4.9.12.4).- The output shall be CW or amplitude MCW, as selected by the operator. The percentage negative peak modulation shall be 10%, minimum.

3.12.12.5 Indicator light period (audio identification) (see 4.9.12.5).- Under either CW or MCW type of transmission the audio "tick" (and indicator lamp) superimposed upon the rf signal shall have a constant "ticking" period of 40 - 120 pulses per minute.

3.12.12.6 Presence of identification on carrier (see 4.9.12.6).- The carrier shall blank at the time the Indicator Lamp flashes.

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3.12.12.7 RF power output (see 4.9.12.7).- The output voltage across the dummy load shall meet the following:

Band	Voltage (min) (rms)
1	25
2	25
3	22
4	20
5	14
6	7

3.12.12.8 Power Supply requirements (see 4.9.12.8).- The Transmitter shall operate from the following power sources and shall not exceed the indicated load currents.

1.5 vdc ± 10% @ 350 ma. ± 10%
67.5 vdc ± 10% @ 4 ma., maximum
90.0 vdc ± 10% @ 20 ma., maximum
135.0 vdc ± 10% @ 20 ma., maximum

(NOTE: The above voltages are as supplied by Batteries BA-404( )/U and BA-424( )/U or equal).

3.13 Direction Finder Set AN/TRD-15(System) (see 4.10).- The Direction Finder Set as a system shall meet the following requirements:

3.13.1 Bearing Sensitivity.- The field intensity required for a threshold bearing shall be no greater than the values listed below:

Frequency (mc)	Field Strength (uv/meter, max)
1.5	9.0
3.0	3.5

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Frequency (mc)	Field Strength (uv/meter, max)
5.0	1.5
7.0	1.5
10.0	1.0
12.0	1.5
15.0	3.0
20.0	3.0

3.13.2 Bearing accuracy.- The indicated bearing shall be within 2 degrees of the true bearing.

3.13.3 Position interaction.- During any electrical test no discernible distortion or bearing error shall be evident on the bearing pattern of one operating position when the Receiver of the other position (Primary or Secondary Console Receiver) is in use.

3.13.4 Zero shift.- The bearing shall not shift more than ±1 degree from the reference bearing at 10 mc when tested at any other frequency in the range of operation (1.5 - 20 mcs).

3.13.5 Stability.- The bearing shift shall not exceed 1/4 degree with line voltage and line frequency variation as per paragraph 3.12.1 in addition to any combination of service conditions as specified in section 3.9.

3.13.6 Phase response.- The bearing error due to phase response of the system shall not exceed the value specified below when the Receiver is detuned off resonance by indicated "Detuning Freq.".

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RECEIVER FREQUENCY		BEARING ERROR
Resonance Frequency (mcs)	Detuning Frequency (cps)	Degrees (Max)
1.5	±750	+2° Total 3°
3.0	±750	+2° Total 3°
5.0	±600	±3/4°
7.0	±500	±3/4°
10.0	±400	±1/2°
15.0	±400	±1/2°
20.0	±200	±1/4°

3.13.7 Two receiver mode operation.- The bearing pattern shall be free from splatter and exhibit no instability due to 1 kc deviation frequency modulation with Receiver operated in two receiver mode.

3.13.8 IF output hum.- The hum level shall be at least 40 db down from referenced signal level observed on an oscilloscope at the RF Detector.

3.14 Final operation (see 4.12).- The AN/TRD-15 (and 1.2 units procured separately) shall operate continuously without failure for 56 hours at room ambient temperature and normal ac line voltage.

3.15 Technical literature, tools and running spare parts.- Technical literature, tools and running spare parts shall be furnished as specified in the contract. (See 6.1).

3.16 Workmanship (see 4.13).- The equipment shall be manufactured and assembled in a thoroughly workmanlike manner.

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4. QUALITY ASSURANCE PROVISIONS

4.1 *Contractor's responsibility.*- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. The supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 *Classification of inspection.*- Inspection shall be classified as follows:

(a) Preproduction (First Article) Inspection (does not include preparation for delivery). (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery.(See 4.14)

4.3 *Preproduction inspection.*- This inspection will be performed by the contractor and witnessed by the Government Representative unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B, and Group C (See Tables II, III, and IV, respectively).

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TABLE I - PREPRODUCTION INSPECTION

Inspection (For additional preproduction inspection see 4.3)	Requirement Paragraph	Inspection Paragraph
Elevation	3.9.3	4.8.3
Vibration (Internal)	3.9.5	4.8.5
System	3.13	4.10.1

4.4 Acceptance inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where Required
Marking	3.5.1
Cleaning	3.5.2
Soldering	3.5.3
Welding	3.5.4
Connections	3.5.5
Grounding	3.5.6
Finish	3.5.7

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4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specification requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government, at its discretion, may perform all or any part of the specified requirements. (See 6.6) Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to Group A, Group B, or Group C inspection shall be preconditioned after final assembly. (See 3.10)

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table II and the ordinary inspection procedures of Specification MIL-STD-105.

TABLE II - GROUP A INSPECTION

Inspection	Requirement Paragraph	Inspection Paragraph	AQL Major	AQL Minor
Construction	3.3	4.15	4%	6.5%
Visual and Mechanical	3.16	4.13	1.5 dphu	6.5 dphu
Electrical				
Amplifier, Electronic Control, AM-3545( )/TRD:			1.5 dphu	All electrical defects are major

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TABLE II - GROUP A INSPECTION (CONTINUED)

Inspection	Requirement Paragraph	Inspection Paragraph	AQL Major	AQL Minor
Tuning Fork Period	3.12.2.1	4.9.2.1		
Synchronism	3.12.2.2	4.9.2.2		
Output Voltage	3.12.2.4	4.9.2.4		
Cable Assy, RF, CG-92G/U (75 FT and 110 FT):	3.12.3	4.9.3	1.0%	
Coupler, Rotary, RF, CU-1096( )/TRD:			1.0%	
Insertion Loss	3.12.4.4	4.9.4.4		
Detector, RF, RF-159( )/TRD:			1.5 dphu	
Loud Speaker	3.12.5.1	4.9.5.1		
AM Limiting	3.12.5.6	4.9.5.6		
Limiting Characteristic	3.12.5.8	4.9.5.8		
Discriminator stability	3.12.5.11	4.9.5.11		

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TABLE II - GROUP A INSPECTION (CONTINUED)

Inspection	Requirement Paragraph	Inspection Paragraph	AQL Major	AQL Minor
Discriminator Linearity	3.12.5.12	4.9.5.12		
Auxilliary Output High	3.12.5.13	4.9.5.13		
Auxilliary Output Low	3.12.5.14	4.9.5.14		
Drive, Goniometer-Rotary Coupler TG-126( )/TRD:			1.0%	
Rotational Synchronism	3.12.6.1	4.9.6.1		
Goniometer Orientation	3.12.6.2	4.9.6.2		
Synchronizing Pulse Orientation	3.12.6.3	4.9.6.3		
Goniometer, Electrical, GO-42( )/TRD:			1.0%	
Bearing Accuracy and Null Ratio	3.12.7.1	4.9.7.1		
Balance Ratio	3.12.7.2	4.9.7.2		
Insertion Loss	3.12.7.3	4.9.7.3		

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TABLE II - GROUP A INSPECTION (CONTINUED)

Inspection	Requirement Paragraph	Inspection Paragraph	AQL Major	AQL Minor
Indicator, Azimuth, IP-669( )/TRD:			1.5 dphu	
Sense and Bearing Display	3.12.8.1	4.9.8.1		
Indicator Pattern Size and Quality	3.12.8.4	4.9.8.4		
Indicator Sensitivity	3.12.8.5	4.9.8.5		
Indicator Accuracy	3.12.8.6	4.9.8.6		
Indicator Alidade and Azimuth Scale Assembly	3.12.8.8	4.9.8.8		
Panel, Power Distribution, SB-1747( )/TRD:			1.0 %	
Output meters	3.12.9.1	4.9.9.1		
Output Voltage Range	3.12.9.2	4.9.9.2		
Overload	3.12.9.3	4.9.9.3		
Power Supply PP-4482( )/TRD:			1.0 %	
B+ Voltage	3.12.10.1	4.9.10.1		

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TABLE II - GROUP A INSPECTION (CONTINUED)

Inspection	Requirement Paragraph	Inspection Paragraph	AQL Major	AQL Minor
Filament Voltage	3.12.10.2	4.9.10.2		
Hum and Noise	3.12.10.3	4.9.10.3		
Receiver, Radio R-725( )/URR:	3.12.11	4.9.11	1.5 dphu	
Transmitter, Radio, T-279( )/UR:			1.5 dphu	
Frequency Range, overlap and calibration accuracy	3.12.12.1	4.9.12.1		
Frequency of Modulation (Period)	3.12.12.3	4.9.12.3		
Depth of Modulation (Negative Peak)	3.12.12.4	4.9.12.4		
Indicator Light Period (Audio Identification)	3.12.12.5	4.9.12.5		
Presence of Identification On Carrier	3.12.12.6	4.9.12.6		
RF Power Output	3.12.12.7	4.9.12.7		
Direction Finder Set AN/TRD-15(System):	3.13	4.10.2	1.0 %	
Final Operation	3.14	4.12	0.65%	

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4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table III and to the special procedures for small-sample inspection of Standard MIL-STD-105. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

TABLE III - GROUP B INSPECTION

Inspection	Requirement Paragraph	Inspection Paragraph	AQL Major	AQL Minor
Shock, Bench Handling:	3.9.6.2	4.8.6.2	4.0%	6.5%
Input Power Requirements:	3.12.1	4.9.1	4.0%	
Coupler, Rotary, RF, CU-1096( )/TRD:			4.0%	
Blending	3.12.4.1	4.9.4.1		
Rotor-Stator Spacing	3.12.4.5	4.9.4.5		
Blending Amplitude Modulation Test	3.12.4.2	4.9.4.2		
Output Isolation	3.12.4.3	4.9.4.3		
Detector, RF, RF-159( )/TRD:			6.5%	
Filament Voltage	3.12.5.2	4.9.5.2		

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TABLE III - GROUP B INSPECTION (CONTINUED)

Inspection	Requirement Paragraph	Inspection Paragraph	AQL Major	AQL Minor
Plate Supply	3.12.5.3	4.9.5.3		
Local Oscillator	3.12.5.4	4.9.5.4		
Receiver Relay	3.12.5.5	4.9.5.5		
Non Symmetrical Limiting	3.12.5.7	4.9.5.7		
High-Low Level Difference	3.12.5.9	4.9.5.9		
5 kc Level	3.12.5.10	4.9.5.10		
Drive, Goniometer-Rotary Coupler TG-126( )/TRD:			4.0%	
Synchronizing Pulse Width	3.12.6.4	4.9.6.4		
Synchronizing Pulse Amplitude	3.12.6.5	4.9.6.5		
Indicator, Azimuth, IP-669( )/TRD:			4.0%	
Indicator High Voltage Supply	3.12.8.2	4.9.8.2		
Indicator Low Voltage Supply	3.12.8.3	4.9.8.3		

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TABLE III - GROUP B INSPECTION (CONTINUED)

Inspection	Requirement Paragraph	Inspection Paragraph	AQL Major	AQL Minor
Cathode-Ray Tube Quadrature	3.12.8.9	4.9.8.9		
Transmitter, Radio, T-279( )/UR:			4.0%	
Frequency Drift	3.12.12.2	4.9.12.2		
Power Supply Requirements	3.12.12.8	4.9.12.8		
Interchangeability:	3.7	4.7	4.0%	

4.5.2.1 Group B sampling plans.- The Group B sampling plans, for the AQL's listed in Table III, shall be as follows:

AQL	Inspection Level
4.0%	S-3
6.5%	S-3

4.5.2.2 Order for inspection within Group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.3 Group C inspection.- This inspection shall be as listed in Table IV and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection. Sample units shall be selected in accordance with paragraph 4.5.3.1.

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TABLE IV - GROUP C INSPECTION

Inspection	Requirement Paragraph	Inspection Paragraph
Temperature	3.9.1	4.8.1
Moisture Resistance	3.9.2	4.8.2
Rain Test	3.9.4	4.8.4
Shock, Bounce	3.9.6.1	4.8.6.1
Electromagnetic Compatability	3.11	4.11
Amplifier, Electronic Control AM-3545( )/TRD:		
Protective Circuit	3.12.2.3	4.9.2.3
Indicator, Azimuth, IP-669( )/TRD:		
Indicator Cathode-Ray Tube Shielding	3.12.8.7	4.9.8.7

4.5.3.1 Sampling for inspection.- Two sample units (excluding preproduction samples) shall be selected at random for Group C inspection in Table IV from the first 50 units produced and from each succeeding 100 units and fraction thereafter produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. Two items are required to permit tests to be conducted at the same time. Both units are not to be subjected to the same test.

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4.5.3.2 Noncompliance.- The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all damage.

4.6 Preconditioning (see 3.10).-

4.6.1 Bounce.- The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

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4.6.2 Electrical.- After final alignment and prior to electrical inspection, the unit shall be operated at nominal rated input voltage for a period of 50 hours continuously. The contractor shall make all necessary repairs and realignments to correct any failure encountered during this 50 hour period. The contractor shall furnish a report of failures encountered to the Contracting Officer's Technical Representative.

4.7 Inspection for interchangeability (see 3.7).- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.7. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

4.8 Service condition tests (see 3.9).- The individual units shall be subjected to the tests in Table V.

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TABLE V - SERVICE CONDITION TABLE

	Temp.	Hum.	Elev.	Rain	Vib.	Bounce	Bench
Antenna	X	X	-	-	-	-	-
Amplifier	X	X	X	-	X	X	X
Cabinet	X	X	-	-	X	-	-
Coupler	X	X	-	-	X	X	X
Detector, RF	X	X	X	-	X	X	X
Drive, Gonio	X	X	-	-	X	X	X
Goniometer	X	X	-	-	X	X	X
Indicator, Azimuth	X	X	X	-	X	X	X
Panel, Power	X	X	-	-	X	X	X
Power Supply	X	X	X	-	X	X	X
Receiver	X	X	-	-	-	-	-
Transmitter	X	X	X	X	X	X	X
System	X	X	-	-	-	-	-

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4.8.1 Temperature test (see 3.9.1).- The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169 except that step 8A, 0° F (+0, -6)° F shall be added as a measurement step. Performance shall be as follows:

Step	Measurement	Performance
1, 3, 8A, 10	Group A Inspection of Table II for items listed in Table V	Full Specification Performance

4.8.2 Moisture-resistance test (see 3.9.2).- The equipment shall be tested as follows:

4.8.2.1 Test conditions.-

a. Do not move equipment from the humidity chamber for measurements.

b. Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.8.2.2 Test method.-

a. Dry at 130° ± 5° F for 24 hours.

b. Condition at 77° ± 5° F and 40 to 50 percent relative humidity for 24 hours.

c. Take measurements as specified in 4.8.2.3 and readjust or realine as necessary to meet full specification requirements.

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d. Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.8.2.3 shall be made during the periods shown on the standard.

e. After cycling has been completed, condition the equipment for 24 hours at 77° ± 5° F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.8.2.3.

4.8.2.3 Performance.- Measurements made during test and specified performance during cycling shall be as listed in 4.8.1.

4.8.2.4 Failure.- If any equipment fails to meet the performance specified in 4.8.2.3 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.8.2.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.8.2.2e, it does not pass the test.

4.8.3 Elevation test (see 3.9.3).- The equipment, placed in its normal operation position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

a. With the equipment placed in the chamber operating and stabilized at room temperature, take the measurements listed in 4.8.1.

b. Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.8.3a. The equipment shall meet specified performance for those measurements.

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c. Remove the power from the equipment and further reduce the barometric pressure to 3.4 ± 0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

d. Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.8.3a. The equipment shall meet specified performance for these measurements.

4.8.4 Rain Test (see 3.9.4).- The transmitter shall be tested as follows:

a. Dry at 150° ± 5° F for 48 hours.

b. Condition for 4 hours at 77° ± 5° F and 40 to 60 percent relative humidity.

c. Take the following measurements to insure that the equipment meets specified performance.

Paragraph	Measurement
4.9.12.1	Frequency Range, etc.
4.9.12.7	Power Output

d. Place the equipment, in its normal operating position, in a test chamber maintained at 77° ± 5° F. Allow simulated rainfall to fall on exposed surfaces of the equipment. The water shall be at a temperature of 65° ± 5° F and shall fall freely at a rate of 2 inches per hour. Maintain this condition for 1 hour.

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e. With the rainfall continuing at the 2-inch rate, introduce a simulated wind blowing horizontally at 20 ± 5 miles per hour against one side of the equipment. The water shall fall on exposed surfaces and be blown by the wind against one side of the equipment. Maintain this condition for 1 hour.

f. Rotate the equipment 90 degrees horizontally, to expose a second side of the component to the wind and rain. Maintain this condition for 1 hour.

g. Continue as in f, until all four sides of the equipment have been exposed to the wind and rain, each side for 1 hour.

h. Immediately after completion of the foregoing, repeat the measurements specified in c without removal from the chamber but with the rain and wind discontinued. Complete the measurements within 1 hour. The equipment shall meet specified performance.

i. Open the equipment. There shall be no evidence of leakage.

4.8.5 Vibration, internal.- Internal vibration of the equipment shall be measured as follows, to determine conformance with 3.9.5:

a. Secure the unit in its rack fixture directly to the vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. The rack mounting fixture shall be a rigid open frame simulating the cabinet or rack into which the unit is normally placed for operation. The unit under test shall be secured to the fixture by its front panel screws and the rear chassis stops (see Drawings SM-B-535902 and SM-C-536043). To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment. The Cabinet shall be rigidly mounted to the vibration table at the four corner mounting holes.

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b. Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch.

c. In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

d. Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.8.6 Shock (see 3.9.6).-

4.8.6.1 Bounce, vehicular (see 3.9.6.1).- The equipment shall be tested on the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawings SC-A-46439. Unless otherwise specified the test shall be as follows:

a. The equipment shall be secured to the 48- by 48-inch vehicular adapter plate using a fixture that will simulate the exact mounting of the equipment in its cabinet (see 4.8.5a) and using appropriate shock mounts as per Specification MS-91527. (The Transmitter shall be strapped to the adapter plate without shock mounts). The adapter plate shall be free of cracks and broken welds, and all unused holes except the 2-inch breather holes shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate.

b. Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

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c. Place the adapter plate on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to safeguard personnel, provided that the fore-and-aft motion of the adapter plate against the backstop is not restrained.

d. Operate the package tester, shafts in phase, for a total of 3 hours at 283 ± 1 rpm. Turn the adapter plate 90 degrees at the end of each 45 minutes, each time in the same direction. Rest periods may be provided to permit cooling of the resilient mounts, provided that the testing time is not less than 3 hours.

4.8.6.2 Bench-handling (see 3.9.6.2).- The chassis-and-front-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid-2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

a. Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

b. Repeat a, with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.9 Detailed requirements (see 3.12).-

4.9.1 Input power tests (see 3.12.1).- Each individual unit and system shall be tested under any combination of input voltage and frequency range specified in 3.12.1. The output voltage switch of the Power Distribution Panel shall be initially adjusted to 115 vac output with 115 vac (or 230 vac) input and thereafter left untouched for this test.

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4.9.2 Amplifier, Electronic Control, AM-3545( )/TRD (see 3.12.2).- The Control Amplifier shall be tested in accordance with all of the procedures in ECOM Drawing SM-B-535949 and the following:

4.9.2.1 Tuning Fork Period (see 3.12.2.1).- The equipment shall be tested in accordance with Drawing SM-B-535949, para. 5.1.4, 5.1.5 and 5.1.6 to determine compliance with requirement 3.12.2.1.

4.9.2.2 Synchronism (see 3.12.2.2).- The equipment shall be tested in accordance with Drawing SM-B-535949, para. 5.1.8.5, 5.1.8.7 and 5.1.8.9 to determine compliance with requirement 3.12.2.2.

4.9.2.3 Protective Circuit (see 3.12.2.3).- With Drive TG-126/TRD operating at synchronous speed,

a. The motor shall be braked to a stop

b. The output leads shall be shorted

c. The output fuse shall be removed.

With conditions returned to normal, the Amplifier shall be restored and shall operate the Drive TG-126/TRD at synchronous speed.

4.9.2.4 Output voltage (see 3.12.2.4).- With the Amplifier operating Drive TG-126/TRD at synchronous speed the output voltage shall meet the requirement of 3.12.2.4.

4.9.3 Cable Assembly, RF, CG-92G/U (75 FT and 110 FT) (see 3.12.3).- Antenna cables shall be tested in accordance with ECOM Drawings SC-D-23249 and SM-B-535747.

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4.9.4 Coupler, Rotary, RF CU-1096( )/TRD (see 3.12.4).- The Rotary Coupler shall be tested in accordance with all of the procedures in ECOM Drawing SM-B-535757 and the following:

4.9.4.1 Blending (see 3.12.4.1).- The equipment shall be tested in accordance with Drawing SM-B-535757, para. 5.1, to determine compliance with requirement 3.12.4.1.

4.9.4.2 Blending Amplitude Modulation (see 3.12.4.2).- The equipment shall be tested in accordance with Drawing SM-B-535757, para. 5.2, to determine compliance with requirement 3.12.4.2.

4.9.4.3 Output Isolation (see 3.12.4.3).- The equipment shall be tested in accordance with Drawing SM-B-535757, para. 5.3, to determine compliance with requirement 3.12.4.3.

4.9.4.4 Insertion loss test (see 3.12.4.4).- The equipment shall be tested in accordance with Drawing SM-B-535757, para. 5.4, to determine compliance with requirement 3.12.4.4.

4.9.4.5 Rotor-Stator spacing (see 3.12.4.5).- Each Rotary Coupler to be subjected to this test shall first be subjected to the following shock:

With the Rotary Coupler bolted to a 2x12x12 inch hardwood block and at rest in its normal operating position, a two pound hammer shall drop from its horizontal position with its pivot 12 inches above the striking point and shall strike the back of the stator plate at the intersection of the center line of the rotor axis. This procedure shall be repeated until the Rotary Coupler has received three blows.

After the hammer blows are applied, there shall be no evidence of binding, increased friction, or increased axial play in the rotor shaft when the shaft is manually operated. The stator plate shall then be removed, the shim thickness and the distance from the plane of the

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rotor face and the plane of the housing flange shall be accurately measured. The difference between the shim thickness and the height of the rotor face above the housing flange shall meet the requirement of 3.12.4.5.

4.9.5 Detector, RF, RF-159( )/TRD (see 3.12.5).- The Detector, RF, RF-159( )/TRD shall be tested in accordance with all of the procedures in ECOM Drawing SM-B-535755 using Power Supply PP-4482( )/TRD and the following:

4.9.5.1 Loud speaker (see 3.12.5.1).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.1.3, to determine compliance with requirement 3.12.5.1.

4.9.5.2 Filament voltage (see 3.12.5.2).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.1.6, to determine compliance with requirement 3.12.5.2.

4.9.5.3 Plate Supply (TP13) (see 3.12.5.3).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.1.8, to determine compliance with requirement 3.12.5.3.

4.9.5.4 Local oscillator (TP1) (see 3.12.5.4).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.1.10, to determine compliance with requirement 3.12.5.4.

4.9.5.5 Receiver relay (K1) (see 3.12.5.5).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.1.11, to determine compliance with requirement 3.12.5.5.

4.9.5.6 AM limiting (TP7) (see 3.12.5.6).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.2.3, to determine compliance with requirement 3.12.5.6

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4.9.5.7 Non Symmetrical Limiting (TP7) (see 3.12.5.7).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.2.4, to determine compliance with requirement 3.12.5.7.

4.9.5.8 Limiting characteristic (TP7) (see 3.12.5.8).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.2.5 and 5.2.6 to determine compliance with requirement 3.12.5.8.

4.9.5.9 High-Low level difference (see 3.12.5.9).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.2.7, to determine compliance with requirement 3.12.5.9.

4.9.5.10 5 kc level (TP9) (see 3.12.5.10).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.2.16, to determine compliance with requirement 3.12.5.10.

4.9.5.11 Discriminator stability (see 3.12.5.11).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.3.5 and 5.3.6 to determine compliance with requirement 3.12.5.11.

4.9.5.12 Discriminator linearity (TP11) (see 3.12.5.12).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.3.12, to determine compliance with requirement 3.12.5.12.

4.9.5.13 Auxilliary output high (TP12) (see 3.12.5.13).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.4.1, to determine compliance with requirement 3.12.5.13.

4.9.5.14 Auxilliary output low (TP12) (see 3.12.5.14).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.4.2, to determine compliance with requirement 3.12.5.14.

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4.9.5.15 Harmonic rejection (J6) (see 3.12.5.15).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.5.4, 5.5.5 and 5.5.6 to determine compliance with requirement 3.12.5.15.

4.9.6 Drive, Goniometer-Rotary Coupler, TG-126( )/TRD (see 3.12.6).- The Drive, Goniometer-Rotary Coupler TG-126( )/TRD shall be tested in accordance with all of the procedures in ECOM Drawing SM-B-536029 and the following.

4.9.6.1 Rotational Synchronism (see 3.12.6.1).- The equipment shall be tested in accordance with Drawing SM-B-536029, para. 5.1, to determine compliance with requirement 3.12.6.1.

4.9.6.2 Goniometer orientation (see 3.12.6.2).- The equipment shall be tested in accordance with Drawing SM-B-536029, para. 5.2, to determine compliance with requirement 3.12.6.2.

4.9.6.3 Synchronizing pulse orientation (see 3.12.6.3).- The equipment shall be tested in accordance with Drawing SM-B-536029, para. 5.3.2, to determine compliance with requirement 3.12.6.3.

4.9.6.4 Synchronizing pulse width (see 3.12.6.4).- The equipment shall be tested in accordance with Drawing SM-B-536029, para. 5.3.2, to determine compliance with requirement 3.12.6.4.

4.9.6.5 Synchronizing pulse amplitude (see 3.12.6.5).- The equipment shall be tested in accordance with Drawing SM-B-536029, para. 5.3.3, to determine compliance with requirement 3.12.6.5.

4.9.7 Goniometer, Electrical, GO-42( )/TRD (see 3.12.7).- The Goniometer, Electrical, GO-42( )/TRD shall be tested in accordance with all of the procedures in ECOM Drawing SM-B-536123 and the following:

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4.9.7.1 Bearing accuracy and null ratio (see 3.12.7.1).- The equipment shall be tested in accordance with Drawing SM-B-536123, para. 5.2, to determine compliance with requirement 3.12.7.1.

4.9.7.2 Balance ratio (see 3.12.7.2).- The equipment shall be tested in accordance with Drawing SM-B-536123, para. 5.3, to determine compliance with requirement 3.12.7.2.

4.9.7.3 Insertion loss (see 3.12.7.3).- The equipment shall be tested in accordance with Drawing SM-B-536123, para. 5.4, to determine compliance with requirement 3.12.7.3.

4.9.8 Indicator, Azimuth IP-669( )/TRD (see 3.12.8).- The Indicator, Azimuth, IP-669( )/TRD shall be tested in accordance with all of the procedures in ECOM Drawing SM-B-535748 and the following:

4.9.8.1 Sense and bearing display (see 3.12.8.1).- The equipment shall be tested in accordance with Drawing SM-B-535748, para 5.4, to determine compliance with requirement 3.12.8.1.

4.9.8.2 Indicator High voltage supply (see 3.12.8.2).- The equipment shall be tested in accordance with Drawing SM-B-535748, para. 5.6, to determine compliance with requirement 3.12.8.2.

4.9.8.3 Indicator Low voltage supply (see 3.12.8.3).- The equipment shall be tested in accordance with Drawing SM-B-535748, para. 5.7, to determine compliance with requirement 3.12.8.3.

4.9.8.4 Indicator pattern size and quality (see 3.12.8.4).- The equipment shall be tested in accordance with Drawing SM-B-535748, para. 5.2, 5.3 and 5.4, to determine compliance with requirement 3.12.8.4.

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4.9.8.5 Indicator sensitivity (see 3.12.8.5).- The equipment shall be tested in accordance with Drawing SM-B-535748, para. 5.8, to determine compliance with requirement 3.12.8.5.

4.9.8.6 Indicator accuracy (see 3.12.8.6).- The equipment shall be tested in accordance with Drawing SM-B-535748, para. 5.10, to determine compliance with requirement 3.12.8.6.

4.9.8.7 Indicator cathode-ray tube shielding (see 3.12.8.7).- During any of the preceding tests external fields, as specified in 3.12.8.7, shall be applied to the CRT shield and the resultant effect on the existing pattern measured for conformance to the requirements of 3.12.8.7.

4.9.8.8 Indicator alidade and azimuth scale assembly (see 3.12.8.8).- The equipment shall be tested in accordance with Drawing SM-B-535748, para. 5.5, to determine compliance with requirement 3.12.8.8.

4.9.8.9 Cathode-Ray Tube Quadrature (see 3.12.8.9).- The equipment shall be tested in accordance with Drawing SM-B-535748, para. 5.5, to determine compliance with requirement 3.12.8.9.

4.9.9 Panel, Power Distribution, SB-1747( )/TRD (see 3.12.9).- The Panel, Power Distribution, SB-1747( )/TRD shall be tested in accordance with all of the procedures in ECOM Drawing SM-B-535948 and the following:

4.9.9.1 Output meter (see 3.12.9.1).- The equipment shall be tested in accordance with Drawing SM-B-535948, para. 5.2, 5.3, 5.7 and 5.8 to determine compliance with requirement 3.12.9.1.

4.9.9.2 Output voltage range (see 3.12.9.2).- The equipment shall be tested in accordance with Drawing SM-B-535948, para. 5.4, 5.6 and 5.10 to determine compliance with requirement 3.12.9.2.

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4.9.9.3 Overload (see 3.12.9.3).- The equipment shall be tested in accordance with Drawing SM-B-535948, para. 5.11 and 5.12 to determine compliance with requirement 3.12.9.3.

4.9.10 Power Supply, PP-4482( )/TRD (see 3.12.10).- The Power Supply, PP-4482( )/TRD shall be tested in accordance with all of the procedures in ECOM Drawing SM-B-535755 and the following:

4.9.10.1 B+ voltage (see 3.12.10.1).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.6.6, 5.6.7, 5.6.8 and 5.6.9 to determine compliance with requirement 3.12.10.1.

4.9.10.2 Filament Voltage (see 3.12.10.2).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.6.11, 5.6.12, 5.6.13 and 5.6.14 to determine compliance with requirement 3.12.10.2.

4.9.10.3 Hum and Noise (see 3.12.10.3).- The equipment shall be tested in accordance with Drawing SM-B-535755, para. 5.6.17, to determine compliance with requirement 3.12.10.3.

4.9.11 Radio Receiver R-725( )/URR (see 3.12.11).- The Radio Receiver R-725( )/URR shall be tested in accordance with all of the procedures in ECOM Drawing SM-B-535758 and the following:

4.9.11.1 IF phase response (see 3.12.11.1).- The equipment shall be tested in accordance with Drawing SM-B-535758, para. 5.1, to determine compliance with requirement 3.12.11.1.

4.9.11.2 IF selectivity (see 3.12.11.2).- The equipment shall be tested in accordance with Drawing SM-B-535758, para. 5.2, to determine compliance with requirement 3.12.11.2.

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4.9.11.3 Beat frequency oscillator leakage (see 3.12.11.3).- The equipment shall be tested in accordance with Drawing SM-B-535758, para. 5.3, to determine compliance with requirement 3.12.11.3.

4.9.11.4 IF output characteristic (see 3.12.11.4).- The equipment shall be tested in accordance with Drawing SM-B-535758, para. 5.4, to determine compliance with requirement 3.12.11.4.

4.9.11.5 AM sensitivity (see 3.12.11.5).- The equipment shall be tested in accordance with Drawing SM-B-535758, para. 5.5, to determine compliance with requirement 3.12.11.5.

4.9.11.6 A.G.C. Characteristics (see 3.12.11.6).- The equipment shall be tested in accordance with Drawing SM-B-535758, para. 5.6, to determine compliance with requirement 3.12.11.6.

4.9.11.7 Overall gain (see 3.12.11.7).- The equipment shall be tested in accordance with Drawing SM-B-535758, para. 5.7, to determine compliance with requirement 3.12.11.7.

4.9.11.8 B.F.O. Tuning Range (see 3.12.11.8).- The equipment shall be tested in accordance with Drawing SM-B-535758, para. 5.8, to determine compliance with requirement 3.12.11.8.

4.9.11.9 IF Frequency (see 3.12.11.9).- The equipment shall be tested in accordance with Drawing SM-B-535758, para. 5.9, to determine compliance with requirement 3.12.11.9.

4.9.11.10 Carrier level meter (see 3.12.11.10).- The equipment shall be tested in accordance with Drawing SM-B-535758, para. 5.10, to determine compliance with requirement 3.12.11.10.

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4.9.12 Transmitter, Radio, T-279( )/UR (see 3.12.12).- The Transmitter, Radio T-279( )/UR shall be tested in accordance with all of the procedures in ECOM Drawing SM-B-536137 and the following:

4.9.12.1 Frequency range, overlap and calibration accuracy (see 3.12.12.1).- The equipment shall be tested in accordance with Drawing SM-B-536137, para. 5.2.7 and 5.2.8, to determine compliance with requirement 3.12.12.1.

4.9.12.2 Frequency drift (see 3.12.12.2).- The equipment shall be tested in accordance with Drawing SM-B-536137, para. 5.2.1, to determine compliance with requirement 3.12.12.2.

4.9.12.3 Frequency of modulation (period) (see 3.12.12.3).- The equipment shall be tested in accordance with Drawing SM-B-536137, para. 5.2.2, to determine compliance with requirement 3.12.12.3.

4.9.12.4 Depth of modulation (negative peak) (see 3.12.12.4).- The equipment shall be tested in accordance with Drawing SM-B-536137, para. 5.2.4, to determine compliance with requirement 3.12.12.4.

4.9.12.5 Indicator light period (audio identification) (see 3.12.12.5).- The equipment shall be tested in accordance with Drawing SM-B-536137, para. 5.2.2, to determine compliance with requirement 3.12.12.5.

4.9.12.6 Presence of identification on carrier (see 3.12.12.6).- The equipment shall be tested in accordance with Drawing SM-B-536137, para. 5.2.3, to determine compliance with requirement 3.12.12.6.

4.9.12.7 R.F. Power output (see 3.12.12.7).- The equipment shall be tested in accordance with Drawing SM-B-536137, para. 5.2.7 and 5.2.8 to determine compliance with requirement 3.12.12.7.

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4.9.12.8 Power Supply requirements (see 3.12.12.8).- The equipment shall be tested in accordance with Drawing SM-B-536137, para. 5.2.6, to determine compliance with requirement 3.12.12.8.

4.10 Direction Finder Set AN/TRD-15 (System) (see 3.13).- The Direction Finding system shall be tested at an interference-free field site provided by the contractor on the preproduction (first article) samples in accordance with 4.10.1 and regular production systems shall be factory tested in accordance with 4.10.2 to meet the requirements of 3.13.

4.10.1 Field test.- A complete Direction Finder Set AN/TRD-15 shall be installed at a test site which is clear of obstructions such as wires, building, towers, power lines, etc. in excess of 300 yards radial distance from the center of the antenna array. The diameter of the antenna array shall be 150 feet. (See Figure 1)

4.10.1.1 Bearing sensitivity (see 3.13.1).- This test shall be performed to determine compliance with 3.13.1.

a. Set up a signal generator, with antenna, at a distance of 300 yards from the center of the d.f. antenna.

b. Adjust the signal generator to transmit an unmodulated carrier, at a high enough level, to produce a steady bearing indication on the Azimuth Indicator.

c. Adjust the D/F Receiver controls for maximum sensitivity at 2 kc i.f. bandwidth.

d. Reduce the signal generator signal level until it is just barely possible to read the bearing. A ± 10 degree bounce shall represent threshold bearing.

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e. Measure the field intensity necessary to produce this threshold bearing, using a field intensity meter located 300 yards ± 1 yard, from the signal generator, and so located away from the antenna system as to prevent interaction.

f. Bearing sensitivity measurements shall be made for the test frequencies 1.5, 3, 5, 7, 10, 12, 15 and 20 mcs following procedure a - e.

4.10.1.2 Bearing accuracy (see 3.13.2).- Local target Bearing Accuracy test shall be performed with a target transmitter employed as a signal source at the test frequencies of 1.5, 3, 5, 7, 10, 12, 15 and 20 mcs. The target transmitter shall be adjusted to produce a strong signal at each test frequency and the d.f. shall be tuned to that frequency. The target transmitter shall be successively placed around the antenna system at 10 degree intervals in a circular path of 300 yards radius from the center of the d.f. antenna system. Bearing shall be taken at each target transmitter setting. The accuracy shall conform to the requirements of 3.13.2.

4.10.1.3 Position interaction (see 3.13.3).- During all test cycles defined in the Bearing Sensitivity and Bearing Accuracy measurements, the Position Interaction tests of the system shall conform to the requirements of 3.13.3.

4.10.1.4 Zero shift (see 3.13.4).- To verify that the indicated bearing does not change as a function of operating frequency, the following test will be performed: As in the Bearing Accuracy test, take a Reference Bearing at 10 mcs. Vary signal generator frequency from 1.5 - 20 mcs and note shift in Reference Bearing. The Zero Shift, defined as shift in Reference Bearing, shall conform to the requirement of 3.13.4.

4.10.1.5 Stability (see 3.13.5).- The stability of the system shall be tested for compliance with 3.13.5 within the system power input range of 3.12.1. Once adjusted for 115 vac output, the power panel output voltage selector switch shall not be readjusted during this test.

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4.10.1.6 Phase response (see 3.13.6).- The phase response characteristic of the system will be measured at test frequencies of 1.5, 3, 5, 7, 10, 15 and 20 mcs for compliance with 3.13.5.

4.10.1.7 Two receiver mode operation (see 3.13.7).- Operate the AN/TRD-15 system in the two receiver mode with both receivers set to 2 kc bandwidth position. Set a radiating FM signal generator to test frequencies of 1.5, 3, 5, 7, 10, 15 and 20 mcs with 1.0 kcs frequency deviation and 400 cps tone modulation (simulation of a frequency shift keyed signal). The indicated bearing shall comply with 3.13.7.

4.10.1.8 IF output hum (see 3.13.8).- The bearing display shall be clear and distinct with no evidence of splits or blurring.

4.10.2 Factory tests.- The Direction Finding System shall be tested in accordance with the procedures in ECOM Drawing SM-B-535759 to meet the requirements of 3.13.

4.10.2.1 Antenna simulator.- The following calibration procedure will be performed on the Antenna Simulator Test Fixture. Details of the Simulator are given in ECOM Drawing SM-B-535759, Figure 1 and Figure 2. The object of the calibration is to correlate the test data taken on the First Article AN/TRD-15 Direction Finder Set at the field site with data taken at the factory.

4.10.2.1.1 Calibration procedure.- The simulator calibration is to be carried out utilizing a strong signal level in order to minimize the effects of noise and interference. For the purpose of this measurement the receiver (R-725/URR) in the Primary Console will be used as a detection device to determine the magnitude of the signal in terms of Diode Load Voltage generated for a calibrated field strength. This same Diode Load Voltage will then be used to calibrate a signal generator when the system is connected to the simulator. The signal generator attenuator can then be calibrated directly in microvolts per meter for each of the test frequencies. In order to obtain the greatest resolution possible, the receiver FUNCTION switch is set to the MGC position and a DC VTVM is connected between the Diode Load output and ground (TB-103 Terminals 14 and 15).

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For example consider the following case:

a. The system is connected to the antenna array as indicated in Figure 1, and the attenuator on the target signal generator is set to +1 dbm. (at the frequency under test).

b. The field strength is measured to be 4 uv/m.

c. The DC voltage output of the Diode Load is measured to be -30 vdc when the FUNCTION switch is in the MGC position.

d. The sytem is now connected to the simulator and the receiver controls are in the same position as in step c,

e. The signal generator is connected to the simulator and the output attenuator adjusted until the VTVM indicates -30 vdc.

f. The attenuator is now read and it indicates an output level of -17 dbm.

g. The level of -17 dbm corresponds to a field strength of 4 uv/m.

h. The simulator may now be used to measure the system sensitivity by reducing the signal generator until a threshold bearing is obtained.

i. The output attenuator of the generator indicates -37 dbm.

j. Then sensitivity in microvolts per meter is equal to a signal 20 db below 4 microvolts or 0.4 microvolts per meter. This relationship is given by:

$$S = \frac{F.S.C.}{\text{Antilog}\ \frac{(\Delta\ db)}{(\ 20\ )}}$$

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S = Sensitivity of Threshold
F.S.C. = Field Strength Calibrated
Δ db = Change in attenuator reading in db for calibration level to threshold value in db.

4.10.2.1.2 Equipment required.- The following equipment will be utilized in performing the simulator calibration measurement:

a. Signal Generator, Hewlett-Packard 606A

b. Master Mobile Citizens Band Antenna Type FG 103 or equal.

c. Field Strength Meter, Ferris, or equal

d. AN/TRD-15 System

e. Portable Motor Generator Set

f. VTVM RCA WV-77E

g. Antenna Simulator

4.10.2.1.3 Detailed test procedure.-

a. Equipment is set up in accordance with Block Diagram shown in Figure 1.

b. Locate the Hewlett-Packard Signal Generator Type 606A and Target Antenna 300 yards from the center Antenna Array.

c. Locate the Ferris Field Strength Meter 200 feet from the center of the Antenna Array and 300 yards from the signal generator.

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d. On the primary receiver set the FUNCTION switch to MGC.

e. Adjust Target Signal Generator attenuator to a level at which the receiver does not overload for a test frequency of 1.5 mcs unmodulated carrier.

f. Use the Ferris Field Strength Meter to measure the field strength for this signal level.

g. Record Diode Load output and Field Strength reading on data sheet.

h. Repeat procedure a - g for test frequencies of 3, 5, 7, 10, 12, 15 and 20 mcs.

i. The equipment is connected to the Antenna Simulator in accordance with ECOM Drawing SM-B-535759, Figure 1. VTVM Diode Load output of Primary Receiver (TB-103 Terminals 14 and 15).

j. Without disturbing the receiver controls set the signal generator output to the level required for Diode Load output voltage obtained in g.

k. Record this attenuator setting on the data sheet.

l. Calculate the signal generator attenuator setting required for the spec threshold limit using the method outlined in para. 4.10.2.1.1 step j.

m. Record this level on data sheet. This level will be used for all factory testing at 1.5 mcs.

n. Repeat steps j - m for the test frequencies in step h. Record these measurements on data sheet. These levels will be used for all factory testing at the other test frequencies.

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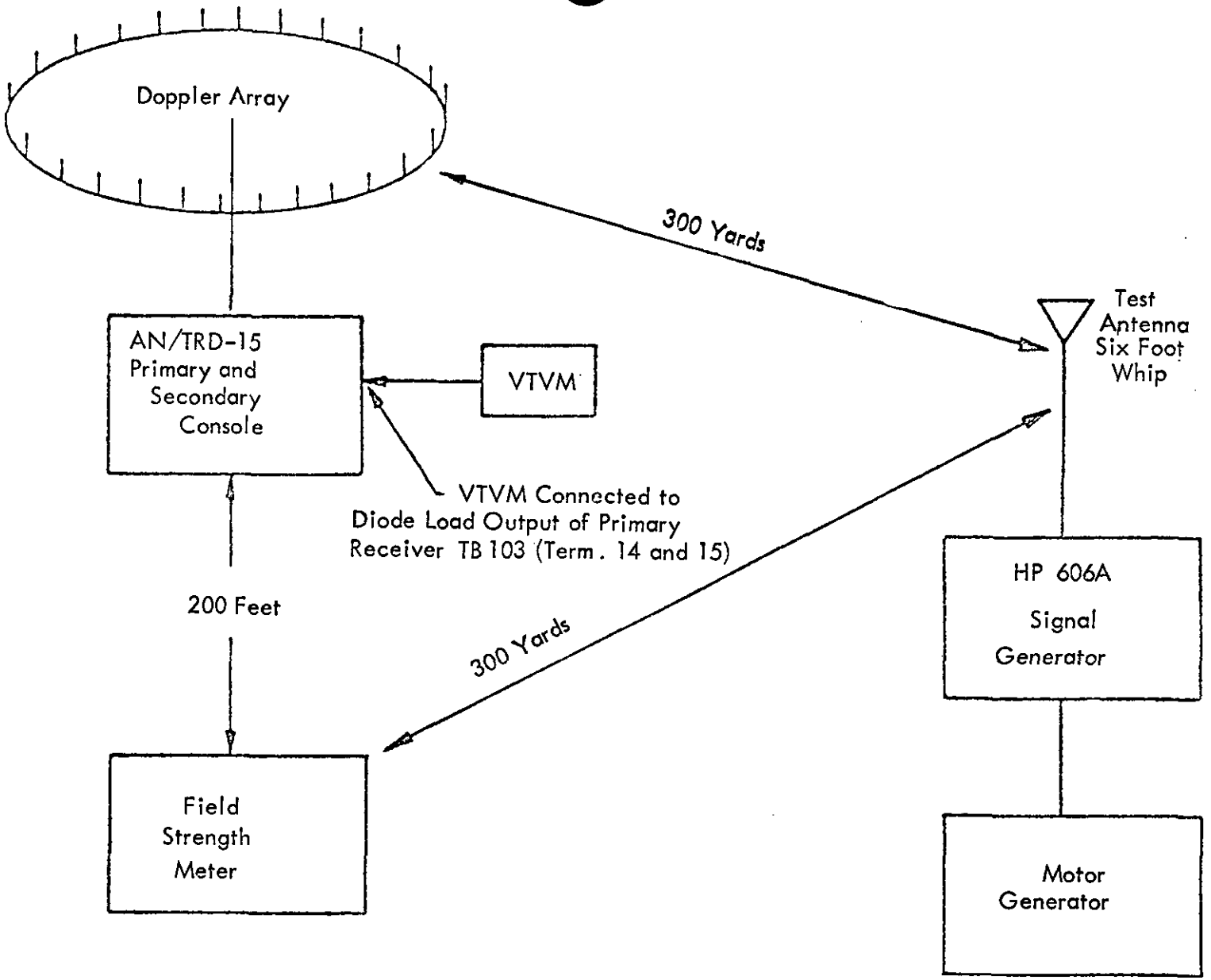


FIGURE 1

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4.11 Electromagnetic compatibility (see 3.11).- The equipment shall be tested for Electromagnetic Compatibility in accordance with Specification MIL-E-55301(EL) and shall be in compliance with requirement 3.11.

4.12 Final operation (see 3.14).- Prior to preparation for delivery, each AN/TRD-15 system shall be subjected to a 56 hour operational test. This test shall consist of the following:

a. Each AN/TRD-15 system (or unit procured separately) shall be operated continuously under normal ambient conditions and nominal input voltage for a period of 56 hours. A target signal, either radiating or simulated shall be supplied to obtain a bearing indication.

b. On each of three days during normal working hours, the system (or unit) shall be shut down and restarted once. All controls available to the operator shall be adjusted or operated including the Power Panel voltage control and the one or two receiver mode switches. When operational checks are completed, the system (or unit) shall remain on and operating.

c. Units which are inoperative at the end of the 56 hour period or which have not operated continuously for the full 56 hours without failure shall not be acceptable. No adjustments, other than those available on the front panels of the units shall be permitted.

4.13 Visual and mechanical inspection.- The units shall be examined for the defects listed in Standard MIL-STD-252.

4.14 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

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4.15 Construction (see 3.3).-

4.15.1 Procedure.- For the purpose of this test, the complete set of drawings for the item listed in 3.3 shall be considered as one lot. Sample drawings shall be selected at random from each lot in accordance with Inspection Level II of MIL-STD-105. Component parts, units or subassemblies corresponding to the sample drawing(s) shall be examined visually, mechanically and electrically (if applicable) against the requirement of the drawing and including any subsidiary specification listed on the drawing. This test shall be performed once, concurrently with the submission of the initial production unit (not preproduction sample) for acceptance inspection or concurrently with the submission of the initial running set of revised production drawings (if applicable), whichever occurs first. Component parts and subassemblies may be examined at accessible points in the production process.

4.15.2 Accept/Reject Criteria.- Acceptance numbers shall be obtained from Table IIA of MIL-STD-105 for the Sample Size Code letter (see 4.15.1) and for the Acceptable Quality Levels (AQL) specified (see 4.5.1). A failure is defined as: 1) Any discrepancy between the part or subassembly and its drawing requirement without regard to the accuracy of the drawing requirement and 2) Any discrepancy between the drawing requirement and other contracted requirements. (Procedures for correcting or revising drawings are covered in the contract documents). Acceptance numbers for resubmitted lots shall be obtained from Table IIB of MIL-STD-105.

4.15.3 Classification of defects.- In addition to the classifications in MIL-STD-252, which concern design parameters, the following shall apply:

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4.15.3.1 Major.-

Electrical parameters incomplete, incorrect or not met.

Mounting dimensions incomplete, incorrect or not met.

Envelope dimensions incomplete, incorrect or not met.

Finish processes incomplete, incorrect or not met.

Mechanical function not specified or not met (torque, clearance, detent, etc.)

Identification marking or color coding missing or incomplete.

Electrical or mechanical adjustment nor specified or not met.

Incorrect or obsolete subsidiary specification cited.

Unauthorized substitution of vendor or part.

Non-Conformance with contract requirements.

Failure to revise drawing when required by Engineering Change.

4.15.3.2 Minor.-

Non-critical dimensions incomplete, incorrect or not met.

Non-compliance with drawing practice.

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4.15.4 Disposition of non-conforming products.- Parts, units and assemblies which do not conform to their construction drawing shall be rejected and corrected prior to resubmission for acceptance. Errors in the construction drawing shall constitute a rejection of the part until the drawing is amended in accordance with the contract procedures for Engineering Changes. Revised drawings shall be furnished, when required, in accordance with the contract provision entitled "Production Drawing Changes".

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be Level A or C, as specified.

5.1.1 Level A.-

5.1.1.1 Cleaning.- Cleaning shall be accomplished in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying.- Drying shall be accomplished in accordance with MIL-P-116.

5.1.1.3 Unit packaging.- Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 and as specified herein.

5.1.1.3.1 Technical literature.- Each technical literature shall be packaged method IC-1.

5.1.1.3.2 Running spares, accessories and ancillary items.- Running spare parts, accessories and ancillary items as required, shall be packaged as follows:

5.1.1.3.2.1 Electron tubes.- All tubes not securely clamped in place shall be removed and packaged individually in accordance with MIL-E-75. Spare tubes shall be packaged in accordance with MIL-E-75.

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5.1.1.3.2.2 Vibrators, fuses, fuselinks, lamps, insulators, and antenna sections.- Each item shall be packaged individually method III as follows: Cushion each item by wrapping in type II paperboard conforming to PPP-P-291. Secure the wrap with Class 1 gummed paper tape conforming to UU-T-111.

5.1.1.3.2.3 Cords and cable.- Each cord and cable shall be packaged individually method III as follows: Wind each item into a coil of proportionate dimensions. Tie in three places with cotton twine, type I, 6 ply, conforming to T-T-871; secure the ends with ties.

5.1.1.3.2.4 All other items.- All other items shall be packaged in accordance with MIL-E-17555.

5.1.1.3.2.5 Stowage within compartments.- When stowage space is provided in the equipment, the applicable items packaged as specified above will be stowed within the compartments provided. Overflow packages, or if no space is provided in the equipment, will be consolidated as specified below for consolidated package.

5.1.1.3.2.6 Consolidated package.- Consolidate the items packaged as specified above within a close-fitting, W5c fiberboard box conforming to PPP-B-636. Fill all voids with type II paperboard conforming to PPP-P-291, to prevent movement. Box closure shall be as specified in the appendix of the box specification.

5.1.1.3.3 Components of Direction Finder Set AN/TRD-15( ).- Packaging shall be developed in accordance with the methods listed below, employing kinds and types of materials which when applied will afford adequate protection against corrosion, deterioration and damage during world-wide shipment, handling and open storage:

Amplifiers	Method II
Receivers	" "

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Transmitters	Method II
Power Panels	" "
Indicator Azimuth	" "
Detector, RF	" "
Power Supply	" "
Goniometer	Method IC
Coupler	" "
Cable Assemblies	" "

The outer container of a method IA-14 or IIb package shall meet the requirements of PPP-B-636, class Weather-resistant. Closure shall be in accordance with the appendix of the box specification. In addition, all seams and joints shall be sealed with pressure-sensitive tape conforming to PPP-T-76.

A humidity indicator card conforming to MS-20003 is required for all method II packages except method IIc.

5.1.2 Level C.- Each components of Direction Finder Set AN/TRD-15 shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing.- Packing shall be level A, B or C as specified.

5.2.1 Level A.- Selection of the containers shall be made from Table I based upon the most economical container required to provide minimum, adequate protection for the packaged item for this packing level of protection.

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5.2.1.1 Container waterproofing.- When the containers being placed directly into the shipping container are not waterproofed, waterproofing shall be effected by use of an individual wrap or a grade B caseliner conforming to MIL-L-10547 and appendix thereto, prior to packing. Dunnage not inclosed within the waterproof wrap or caseliner shall consist of materials treated to resist water.

TABLE VI - CONTAINER REQUIREMENTS

Container	Specification	For Level A	For Level B	Max. Gross Wt. Limitation (lbs)
Boxes, Wood, Nailed and Lock-Corner, Style 4	PPP-B-621	Class 2	Class 1	200 (Note A)
Boxes, Wood, Nailed and Lock-Corner, Style 7	PPP-B-621	Class 2	Class 1	1000 (Note B)
Box, Wirebound, Style 2 or 3	PPP-B-585	Class 3	Class 1	200
Box, Wood Cleated Fiberboard	PPP-B-591	Class II	Class 1	200 (Note A)
Box, Cleated Plywood	PPP-B-601	Overseas Type	Domestic Type	200 (Note A)
Boxes, Fiberboard, Corrugated, Triple Wall, Style E	PPP-B-640	Class 2	Class 1	200 (Note A)

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TABLE VI - CONTAINER REQUIREMENTS (CONTINUED)

Container	Specification	For Level A	For Level B	Max. Gross Wt. Limitation (lbs)
Box, Fiberboard, Type CF	PPP-B-636	Class Weather-resistant	Class Domestic	(Note C)
Box, Fiberboard, Corrugated, Double Wall, Weather Resistant	MIL-B-43291	Type CF	-	Table II

NOTE A: When the weight of a single item together with the shipping container exceeds 200 pounds, the maximum gross weight limitation of the container specification shall apply. In addition, skids, or pallet bases shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment.

NOTE B: Style 7 container conforming to PPP-B-621 shall be considered for use only when the weight of a single item exceeds 400 pounds. Sheathed crates conforming to MIL-C-104 shall be used only when the weight of a single item together with the shipping container exceeds 1000 pounds.

NOTE C: The maximum gross weight limitations for Special Requirements as cited in PPP-B-636 shall apply.

Wraps or caseliners may be omitted from fiberboard boxes provided all seams, corners and manufacturer's joint are waterproofed by sealing with tape, not less than 2 inches wide, conforming to PPP-T-76. The tape shall be centered over the seams and joints and shall

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extend over all corners and edges of the box a minimum of two inches onto the adjacent box panels. Tape shall be applied over the lengthwise seams of the outer flaps closing the openings of the box and over the manufacturer's joint prior to tape being applied to the edge seams of the box. The tape applied to the manufacturer's joint shall only fully cover the joint and not be extended over the corners of the box onto the adjacent panels.

5.2.1.2 Container closure.- Closure shall be in accordance with the applicable container specification or appendix thereto.

5.2.1.3 Strapping and reinforcing.- Metal strapping conforming to QQ-S-781, type I, class B shall be applied to wood and wood-cleated shipping containers in accordance with the requirements of the applicable box specification or appendix thereto. Fiberboard shipping containers shall be reinforced by pressure sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level B.- Shall be as specified in 5.2.1 except that Level B containers of Table VI shall apply and 5.2.1.1 shall not apply.

5.2.3 Level C.- Components of Direction Finder AN/TRD-15( ) shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Type required.

c. Level of packaging required for shipment (Level A, B or C, See Section 5).

d. The specific paragraphs of Section 5 which are applicable to the particular procurement.

e. Production inspection:

(1) Two sample units of each item cited in Section 1.2 are generally required so that lengthy environmental and reliability tests can be completed within the contractually specified time period.

(2) Preproduction pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

f. Marking and shipping of samples.

g. Place of final inspection.

h. Technical literature, quantity of tools and running spare parts required (see 3.15).

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6.2 Color.-

6.2.1 Paint finish.- Samples of color chips for colors specified in the subsidiary documents (see 3.3) may be obtained upon request to Commanding Officer, U. S. Naval Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania.

6.2.2 Textiles.- Standard samples of Olive Drab No. 7 may be obtained from the Sample Loan Unit, Military Clothing and Textile Supply Agency, Philadelphia, Pennsylvania. When requesting samples, the type of material should be specified.

6.3 Definition of Level B packaging.- Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.4 Location of operational inspection.- It is desirable that the operational inspection (4.12) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.5 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/TRD-15W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.6 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provision established in this specification and contract. The amount of verification inspection by the Government will be adjusted

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to make maximum utilization of the contractor's quality control system and the quality history of the product.

CUSTODIAN
Army (EL)

PREPARING ACTIVITY
Army (EL)

PROJECT NUMBER
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